

Technical Document #55: Software Engineering - Comprehensive Overview

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Code review examines code changes before merging. It improves code quality, catches bugs, and shares knowledge across the team.

Test-driven development (TDD) writes tests before writing code. It improves code quality and design through small, incremental changes.

Technical debt represents the cost of choosing easy solutions over better ones. It accumulates over time and slows development velocity.

Microservices architecture structures applications as independently deployable services. Each service owns its data and business logic.

Autonomous vehicles use a combination of sensors, AI, and mapping to navigate without human intervention. Self-driving technology is rapidly advancing.

Sustainability and environmental responsibility are becoming key considerations in technology design. Green computing aims to reduce the environmental impact of technology.

The Internet of Things (IoT) connects billions of devices worldwide. This connectivity generates massive amounts of data that can be analyzed for insights.

Key Takeaways:

- Microservices architecture structures applications as independently deployable services.
- Test-driven development (TDD) writes tests before writing code.
- Design patterns provide reusable solutions to common software design problems.